Exact Orthogonal-CI Contraction for Arbitrary Selected Valence-Bond Structure Spaces

Theory and implementation note

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Abstract

An arbitrary selected valence-bond (VB) structure space is generally not invariant under orbital orthogonalization. This does not prevent exact evaluation of its Hamiltonian and overlap actions through an orthogonal configuration-interaction (CI) representation. The transformed wave function must be retained, but its complete spin-string product array need not be formed. We derive an exact source-factor representation, apply the Slater–Condon one-body generators separately to the two spin factors, and contract only the original products needed for the final structure-space projection. The resulting equations preserve the selected variational space without a closure assumption or intermediate amplitude truncation. Source and channel tiling bound the numerical workspace. Optional spin-exchange reduction and causal pruning of the exterior adjoint are derived separately. The scope is the Hamiltonian/overlap action, its active-integral adjoint, and the factor representation of the directional integral adjoint.

1 The selected VB space

Let n be the number of active spatial orbitals, with n_α and n_β active electrons of each spin. The complete one-spin string spaces have dimensions

$$U_\alpha = \binom{n}{n_\alpha}, \qquad U_\beta = \binom{n}{n_\beta}. \tag{1}$$

An index a identifies an α string and b a β string. For a selected VB structure I, let $w_{ab,I}$ be its signed expansion coefficient on the corresponding spin-string product, with all normalization conventions included. For structure coefficients $\mathbf{C}$, define

$$X = \mathcal{W}\mathbf{C}, \qquad X_{ab} = \sum_I w_{ab,I} C_I. \tag{2}$$

Here $\mathcal{W}$ is a sparse linear map from structure coefficients to a matrix of spin-string product coefficients. The matrix notation does not prescribe a dense allocation: the map can be stored as sparse incidences between structures and touched string products. It introduces no determinant-pair traversal or separately optimized determinant coefficient vector.

The original support and its two marginal index sets are

$$\Omega = \{(a,b) : w_{ab,I} \neq 0 \text{ for at least one selected } I\}, \qquad t = |\Omega|, \tag{3}$$

$$A_\Omega = \{a : \exists b,\ (a,b) \in \Omega\}, \qquad J = \{b : \exists a,\ (a,b) \in \Omega\}, \qquad r = |J|. \tag{4}$$

Support is determined by the selected structures, not by numerical smallness or cancellations in a particular Davidson vector. Using the Frobenius inner product, the adjoint structure map is

$$(\mathcal{W}^* Z)_I = \sum_{(a,b)\in\Omega} w_{ab,I} Z_{ab}. \tag{5}$$

Thus only entries in Ω are needed at the *end* of a complete action. Nothing in this statement permits dropping intermediate entries outside Ω.

2 Orthogonalization without a closure assumption

At a fixed orbital point, write the nonorthogonal active orbitals as $\phi = \chi R$, where χ is an orthonormal basis of the same active one-particle space. For example, if O is their positive-definite overlap,

$$O = R^{\mathsf{T}} R, \qquad \chi = \phi R^{-1}. \tag{6}$$

This is an evaluation-coordinate change, not a change to the physical localization constraints or to the orbital optimization variables. Multilinearity of antisymmetrized products gives the exterior maps

$$T_\sigma = \wedge^{n_\sigma} R, \qquad (T_\sigma)_{ai} = \det R[a, i], \qquad \sigma \in \{\alpha, \beta\}, \tag{7}$$

where the row and column subsets contain n_σ orbitals with a fixed ordering convention. The orthogonal-basis coefficient matrix is

$$Y = \mathcal{T} X = T_\alpha X T_\beta^{\mathsf{T}}, \qquad \mathcal{T}^* Z = T_\alpha^{\mathsf{T}} Z T_\beta. \tag{8}$$

The minor matrices T_σ are mathematical definitions; the implementation applies elementary exterior transforms without assembling them densely.

Let $\mathcal{H}$ be the active Hamiltonian action in the orthogonal spin-string product basis. The selected structure Hamiltonian and overlap are exactly

$$\boxed{H = \mathcal{W}^* \mathcal{T}^* \mathcal{H} \mathcal{T} \mathcal{W}, \qquad M = \mathcal{W}^* \mathcal{T}^* \mathcal{T} \mathcal{W}.} \tag{9}$$

An orbital-dependent scalar inactive-plus-nuclear energy E_0 contributes $E_0 M$ to the total structure Hamiltonian and is separate from the active action H in this note. The variational energy is

$$E = E_0 + \frac{\mathbf{C}^{\mathsf{T}} H \mathbf{C}}{\mathbf{C}^{\mathsf{T}} M \mathbf{C}}, \tag{10}$$

with H denoting the active contribution of Eq. (9).

Proposition 1 (Exact selected-space restriction)**.** *For any selected structure map $\mathcal{W}$, Eq.* (9) *gives the exact selected-space Hamiltonian and overlap, whether or not $\mathcal{T}$* $\operatorname{ran} \mathcal{W} \subseteq \operatorname{ran} \mathcal{W}$.

Proof. For arbitrary structure vectors $\mathbf{u}$ and $\mathbf{v}$, the two physical states have orthogonal-basis amplitudes $\mathcal{T}\mathcal{W}\mathbf{u}$ and $\mathcal{T}\mathcal{W}\mathbf{v}$. Their Hamiltonian matrix element is

$$\langle \mathcal{T}\mathcal{W}\mathbf{u}, \mathcal{H}(\mathcal{T}\mathcal{W}\mathbf{v}) \rangle_{\mathrm{F}} = \mathbf{u}^{\mathsf{T}} \big(\mathcal{W}^* \mathcal{T}^* \mathcal{H} \mathcal{T} \mathcal{W} \big) \mathbf{v}. \tag{11}$$

Replacing $\mathcal{H}$ by the identity gives the overlap. No intermediate restriction to $\operatorname{ran} \mathcal{W}$ was inserted. □

The nonclosure issue arises only if a different operation is used, for example by zeroing transformed amplitudes outside the original selected support before applying $\mathcal{H}$. That operation changes the wave function. Equation (9) does not do so. Davidson vectors and Ritz coefficients remain in the selected VB structure space; the orthogonal representation is used internally to evaluate their images. For linearly independent selected structures, their eigenproblem remains

$$H\mathbf{C} = E_{\mathrm{act}} M \mathbf{C}. \tag{12}$$

The adjoint $\mathcal{W}^*$ contracts against the selected test structures; it is not an inverse coefficient transform. The nonidentity overlap M is retained, so this procedure does not replace the selected VB problem by an unconstrained orthogonal-CI eigenproblem.

2.1 A minimal nonclosed example

Consider a two-by-two coefficient matrix with the single selected product $X = e_2 e_2^{\mathsf{T}}$, and let

$$T_\alpha = \begin{pmatrix} 1 & a \\ 0 & 1 \end{pmatrix}, \qquad T_\beta = \begin{pmatrix} 1 & b \\ 0 & 1 \end{pmatrix}. \tag{13}$$

This is a linear-algebra example of the coefficient map, not an assertion that a single spin product is a spin-adapted VB structure. Its transformed amplitudes are

$$Y = \begin{pmatrix} ab & a \\ b & 1 \end{pmatrix} = \begin{pmatrix} a \\ 1 \end{pmatrix} \begin{pmatrix} b \\ 1 \end{pmatrix}^{\mathsf{T}}. \tag{14}$$

For nonzero a and b, all four products are present, so the original one-dimensional selected space is not closed. Nevertheless, two vectors represent the complete Y exactly. The overlap is $(1 + a^2)(1 + b^2)$, not the value 1 obtained by retaining only Y_{22}. An exact Hamiltonian contraction must likewise use all transformed amplitudes, explicitly or implicitly. The factorized route below retains them implicitly.

3 Exact source factors

3.1 Column decomposition

Order the original touched β columns as $J = (j_1, \ldots, j_r)$, and set

$$A_{:\ell} = X_{:j_\ell}, \qquad B_{:\ell} = e_{j_\ell}, \qquad X = AB^{\mathsf{T}}. \tag{15}$$

This is an identity for every input vector. It is not a truncated singular value decomposition, and r is not a fitted rank. Transforming only the two factors gives

$$L = T_\alpha A, \qquad F = T_\beta B, \qquad \mathcal{T}X = LF^{\mathsf{T}}. \tag{16}$$

The factors have dimensions $U_\alpha \times r$ and $U_\beta \times r$. The complete transformed $U_\alpha \times U_\beta$ matrix need not be formed, even if every one of its entries is nonzero.

The distinction between r and the number of structures matters. A single covalent structure with m disjoint singlet pairs generally touches 2^m original source columns. Several structures can share most of those columns, and a broad selected space can touch all U_β columns. Thus rank $X \le r$ is an upper bound, not an identification of r with one, with rank X, or with the number of structures.

3.2 A minimum mixed row/column cover

Grouping every source on the β side is not always economical. View the original input support as a bipartite graph: vertices are the touched α and β strings, and each supported product (a, b) is one edge. Choose a vertex cover $C_\alpha \cup C_\beta$, so that every edge touches at least one chosen vertex. Assign an edge to its α endpoint if that endpoint is in C_α; otherwise assign it to its covered β endpoint. With this disjoint assignment,

$$X = \sum_{a \in C_\alpha} e_a u_a^{\mathsf{T}} + \sum_{b \in C_\beta} v_b e_b^{\mathsf{T}} = AB^{\mathsf{T}}. \tag{17}$$

Here $(u_a)_b = X_{ab}$ only for edges assigned to row a, and $(v_b)_a = X_{ab}$ only for edges assigned to column b; the remaining entries are zero. Thus no coefficient is counted twice, including edges whose two endpoints are both covered. Each chosen vertex defines one coordinate-anchored outer product.

Let $\nu(\Omega)$ be the maximum matching size, i.e., the largest number of edges with no shared endpoints. A minimum cover has cardinality

$$r_* = |C_\alpha| + |C_\beta| = \nu(\Omega) \le \min(|A_\Omega|, |J|). \tag{18}$$

For completeness, the cover can be certified directly from a maximum matching. Start at unmatched α vertices and follow alternating unmatched/matched edges. If the reached vertex sets are Z_α and Z_β, then $(A_\Omega \setminus Z_\alpha) \cup Z_\beta$ covers every edge and contains exactly one endpoint per matched edge. Any cover must contain at least one endpoint of each disjoint matched edge, establishing minimality. The implementation uses layered augmenting paths, with $O(t\sqrt{U_\alpha + U_\beta})$ worst-case planning work and linear graph storage. The plan is independent of the numerical Davidson vector.

This is a minimal factor count for *support-only coordinate-anchored* decompositions, not a numerical rank claim. In particular, rank $X \le \nu(\Omega)$; additional coefficient relations can lower the actual rank further. A diagonal support needs one source per edge. In contrast, the support

$$\{(a_0, b_j) : 1 \le j \le m\} \cup \{(a_i, b_0) : 1 \le i \le m\} \tag{19}$$

needs only two mixed sources, although either a pure row or pure column decomposition needs $m + 1$ sources. This reduces the factor dimension, rather than enlarging a tile or imposing an amplitude threshold.

All subsequent Hamiltonian and sampling equations apply unchanged, with $r = r_*$ and the factors of Eq. (17). When a common exchange parity is used, construct the graph from the *representative input* support of Section 6.1; the final projection still uses the complete original Ω. If the original column cover is already minimal, its ordering may be retained to avoid changing established summation order.

4 Hamiltonian action on the factors

4.1 One-body channels and the complete active Hamiltonian

Use real spatial orbitals and real, spin-independent molecular integrals. Let $e_{pq} = \sum_\sigma a^\dagger_{p\sigma} a_{q\sigma}$, and use the chemists' notation $(pq|rs)$ for two-electron integrals. The active Hamiltonian is

$$\hat{H} = \sum_{pq} h_{pq} e_{pq} + \frac{1}{2} \sum_{pqrs} (pq|rs)\big(e_{pq} e_{rs} - \delta_{qr} e_{ps}\big). \tag{20}$$

Here h includes the effective one-electron contribution of the inactive orbitals. Define

$$f_{ps} = h_{ps} - \frac{1}{2} \sum_q (pq|qs), \qquad N = n_\alpha + n_\beta > 0. \tag{21}$$

Packed pair labels $P = (p, q)$ have $p \ge q$, so there are $d = n(n+1)/2$ channels. Define the symmetric generators

$$\hat{E}_{\sigma P} = \begin{cases} a^\dagger_{p\sigma} a_{p\sigma}, & p = q, \\ a^\dagger_{p\sigma} a_{q\sigma} + a^\dagger_{q\sigma} a_{p\sigma}, & p > q, \end{cases} \qquad \hat{E}_P = \hat{E}_{\alpha P} + \hat{E}_{\beta P}. \tag{22}$$

Let $f_P = f_{pq}$, $V_{PQ} = (pq|rs)$ for $Q = (r, s)$, and $d_P = 1$ for diagonal pairs and zero otherwise. A symmetric kernel containing the *entire* active Hamiltonian is

$$K_{PQ} = \frac{1}{2} V_{PQ} + \frac{f_P d_Q + d_P f_Q}{2N}, \qquad \hat{H} = \sum_{PQ} K_{PQ} \hat{E}_P \hat{E}_Q. \tag{23}$$

Indeed, $\sum_P d_P \hat{E}_P = \hat{N} = N\hat{I}$ on the fixed-electron space. The second term in K therefore gives $\sum_P f_P \hat{E}_P$; the first term supplies the quadratic part of Eq. (20). The usual integral permutation symmetries account for both ordered off-diagonal orbital pairs through $\hat{E}_P$; no extra off-diagonal multiplicity is needed in K.

Let $E_{\sigma P}$ be the matrix of $\hat{E}_{\sigma P}$ on its one-spin string space. The Slater–Condon rules make this operation a sparse signed occupation/replacement scatter, rather than a dense string pair contraction. Its action on a product matrix is

$$\mathcal{E}_P(Z) = E_{\alpha P} Z + Z E_{\beta P}^{\mathsf{T}}, \qquad \mathcal{H} = \sum_{PQ} K_{PQ} \mathcal{E}_P \mathcal{E}_Q. \tag{24}$$

These identities require a complete carrier for each fixed-spin sector, or an exactly equivalent representation of all required intermediates. Arbitrary *selected VB structure spaces* need not be complete. The two statements are different.

4.2 Separation of the two spin factors

Define the same-spin channel actions

$$D_\sigma = \sum_{PQ} K_{PQ} E_{\sigma P} E_{\sigma Q}. \tag{25}$$

They include the particular distribution of one-electron terms implied by Eq. (23); they are not independently assigned physical α- and β-spin Hamiltonians. Expanding the two generators gives

$$\begin{aligned} \mathcal{E}_P \mathcal{E}_Q (L F^{\mathsf{T}}) = {} & (E_{\alpha P} E_{\alpha Q} L) F^{\mathsf{T}} + L (E_{\beta P} E_{\beta Q} F)^{\mathsf{T}} \\ & + (E_{\alpha P} L)(E_{\beta Q} F)^{\mathsf{T}} + (E_{\alpha Q} L)(E_{\beta P} F)^{\mathsf{T}}. \end{aligned} \tag{26}$$

The last two terms become identical after summation because $K_{PQ} = K_{QP}$. Consequently,

$$\boxed{\mathcal{H}(L F^{\mathsf{T}}) = (D_\alpha L) F^{\mathsf{T}} + L (D_\beta F)^{\mathsf{T}} + 2 \sum_{PQ} K_{PQ} (E_{\alpha P} L)(E_{\beta Q} F)^{\mathsf{T}}.} \tag{27}$$

The proof uses commutativity of different spin sectors, not commutativity of generators within one spin sector. It applies to unequal U_α, U_β, unequal electron counts, ionic structures, and arbitrary mixtures of selected structures.

5 Direct sampling and structure projection

Introduce the pulled-back factors

$$L_0 = T_\alpha^{\mathsf{T}} L, \qquad F_0 = T_\beta^{\mathsf{T}} F, \tag{28}$$

$$L_H = T_\alpha^{\mathsf{T}} D_\alpha L, \qquad F_H = T_\beta^{\mathsf{T}} D_\beta F, \tag{29}$$

$$L_P = T_\alpha^{\mathsf{T}} E_{\alpha P} L, \qquad G_P = T_\beta^{\mathsf{T}} \sum_Q K_{PQ} E_{\beta Q} F. \tag{30}$$

Combining Eqs. (8) and (27), the needed original product entries are

$$m_{ab} = \sum_{\ell=1}^{r} (L_0)_{a\ell} (F_0)_{b\ell}, \tag{31}$$

$$h_{ab} = \sum_{\ell=1}^{r} \left[(L_H)_{a\ell} (F_0)_{b\ell} + (L_0)_{a\ell} (F_H)_{b\ell} + 2 \sum_P (L_P)_{a\ell} (G_P)_{b\ell} \right], \qquad (a,b) \in \Omega. \tag{32}$$

For any Davidson input $\mathbf{v}$, build $X = \mathcal{W}\mathbf{v}$ and return

$$(M\mathbf{v})_I = \sum_{(a,b)\in\Omega} w_{ab,I} m_{ab}, \qquad (H\mathbf{v})_I = \sum_{(a,b)\in\Omega} w_{ab,I} h_{ab}. \tag{33}$$

Proposition 2 (Exact factored selected-space action)**.** *Equations* (15)–(33) *evaluate Eq.* (9) *exactly for arbitrary selected structures, without constructing a complete transformed product array.*

Proof. Equation (15) is the exact decomposition into nonzero original columns. Linearity gives Eq. (16), and expansion of the Hamiltonian gives Eq. (27). Applying $\mathcal{T}^*$ to each outer product yields Eqs. (31) and (32). All other final entries have zero coefficient in $\mathcal{W}^*$ by Eq. (5). The final sampling therefore omits no contribution to Eq. (9). □

The contraction can be read as the following sequence:

$$\mathbf{v} \xrightarrow{\mathcal{W}} X = AB^{\mathsf{T}} \xrightarrow{T_\alpha, T_\beta} LF^{\mathsf{T}} \xrightarrow{\text{spin generators}} \text{factor images} \xrightarrow{T_\alpha^{\mathsf{T}}, T_\beta^{\mathsf{T}}} (h, m)|_\Omega \xrightarrow{\mathcal{W}^*} (H\mathbf{v}, M\mathbf{v}). \quad (34)$$

The middle objects are represented by factors and channel panels, not by a selected-space approximation to the transformed wave function. This is how Slater–Condon evaluation remains available without requiring selected structures to form an invariant space.

6 Two additional exact reductions

6.1 Common spin-exchange parity

An optional reduction is available when the two one-spin bases and their orbital maps coincide, and every selected structure has the same certified transpose parity $\eta = \pm 1$. Define

$$\Pi X = X^{\mathsf{T}}, \qquad P_\eta = \tfrac{1}{2}(\mathcal{I} + \eta\Pi), \qquad \Pi\mathcal{W} = \eta\mathcal{W}. \quad (35)$$

Construct X_{rep} by keeping $a < b$ with coefficient $2X_{ab}$, keeping diagonal entries with coefficient X_{aa} for $\eta = +1$, and setting the other entries to zero. Odd parity has zero diagonal. Then

$$P_\eta X_{\text{rep}} = X. \quad (36)$$

Since $\mathcal{T}$ and a spin-independent $\mathcal{H}$ commute with exchange, their original-basis composite $\mathcal{A} = \mathcal{T}^*\mathcal{H}\mathcal{T}$ also commutes. Furthermore $\mathcal{W}^* P_\eta = \mathcal{W}^*$. It follows that

$$\mathcal{W}^*\mathcal{A}X_{\text{rep}} = \mathcal{W}^* P_\eta \mathcal{A} X_{\text{rep}} = \mathcal{W}^*\mathcal{A}P_\eta X_{\text{rep}} = \mathcal{W}^*\mathcal{A}X. \quad (37)$$

The same proof applies to $\mathcal{T}^*\mathcal{T}$ for the overlap. Source factorization may therefore use X_{rep}, but the final output sampling still uses *both* halves of Ω. The unprojected image of X_{rep} is not itself a parity eigenvector.

For a single structure with m independent covalent pairs, this can reduce the source count from 2^m to 2^{m-1}. It does not imply that the touched column count always halves for an arbitrary selected space. Equal spin dimensions alone are insufficient; common parity must be checked from the structure map. Parity is not a substitute for total-spin adaptation. Without it, the unreduced factorization remains exact.

6.2 Causal rows of the exterior adjoint

Only A_Ω rows of the α pullbacks and J rows of the β pullbacks enter Eq. (31). An elementary forward exterior shear updates $z_j \leftarrow z_j + s\epsilon z_i$, with orbital coefficient s and fermionic sign $\epsilon = \pm 1$. Its adjoint updates $z_i \leftarrow z_i + s\epsilon z_j$. Starting from the requested final rows, traverse the adjoint operations in reverse chronological order. Retain an update exactly when its destination is needed, and then mark its source as needed. Execute the retained updates in their original adjoint order, including the diagonal exterior scaling.

For a row selector Q_A, this computes $Q_A T_\sigma^{\mathsf{T}} Z$ exactly. Every intermediate row that can influence the answer is retained. The pruning concerns paths through a linear transformation, not closure of the selected VB space and not the magnitude of an amplitude. A bitmap on one-spin rows and retained shear-edge indices are sufficient; no product array or dense compound matrix is required.

7 Tiling, storage, and arithmetic

7.1 Streaming the exact equations

Split the r source columns into panels of width s, and the d channels into panels of width p. Different source columns are summed independently in Eq. (31); the Hamiltonian action never couples source column labels to one another. For each source panel:

1. Build its sparse A, B, and apply T_α, T_β to obtain L, F.
2. For a channel panel $\mathcal{P}$, scatter $E_{\sigma P}$ on the corresponding factors.
3. Stream channel panels Q to accumulate $V_{\sigma P} = \sum_Q K_{PQ} E_{\sigma Q} F_\sigma$, where $F_\alpha = L$ and $F_\beta = F$. This is a channel GEMM on flattened spin-row/source-column panels.
4. Gather $\sum_{P\in\mathcal{P}} E_{\sigma P} V_{\sigma P}$ into the same-spin images. Pull back $E_{\alpha P} L$ and $V_{\beta P}$ and add the cross-spin dot products on Ω.
5. After all channel panels, pull back the base and same-spin factors, add their samples, and reuse the slabs for the next source panel.

Finally apply $\mathcal{W}^*$ once to the accumulated support arrays. Source, channel, and right-hand-side loops reuse shared backing allocations; threads do not receive private copies of the carrier.

The explicit source/channel numerical slabs require

$$O\big((U_\alpha + U_\beta) s p\big) \tag{38}$$

storage, plus $O(t)$ support coefficients and images per processed vector. Additional persistent storage includes the sparse structure map, one-spin link/exterior metadata, and the $d \times d$ kernel K. The source slabs do not include any $U_\alpha U_\beta$ coefficient array. This is a workspace statement, not a bound on the total process memory. For an overlap-only action no channel panels or same-spin Hamiltonian images are required, and the factor workspace reduces to $O((U_\alpha + U_\beta)s)$.

7.2 Where a speed advantage is obtained

Let q_σ count elementary exterior shear edges. With a full channel panel ($p = d$), a leading-work bound per input vector is

$$\begin{aligned} O\Big(& r(U_\alpha + U_\beta) d^2 + r d (q_\alpha + q_\beta) + t r d \\ & + r \sum_\sigma U_\sigma n_\sigma (n - n_\sigma + 1) + \mathrm{nnz}(\mathcal{W}) \Big). \end{aligned} \tag{39}$$

The terms describe channel mixing, exterior pullbacks, final sampling, Slater–Condon spin links, and the initial/final structure contractions. The displayed shear count is an upper bound; causal pruning reduces the selected-row pullbacks. Forward base-factor transforms remain necessary. For $p < d$, channel mixing keeps the same leading arithmetic order, while spin scatters can repeat across channel panels by a factor up to $\lceil d/p \rceil$. A smaller workspace does not change any equation or variational space.

A generated-product contraction instead has channel-mixing work of order $N_{\mathrm{cl}} d^2$, where N_{cl} counts the generated product intermediates needed by that contraction order. When this support is broad and $r \ll \min(U_\alpha, U_\beta)$, the factorized order replaces the product dimension by a source-weighted sum of one-spin dimensions. For $U_\alpha \simeq U_\beta = U$ and $N_{\mathrm{cl}} \simeq U^2$, the leading mixing ratio is approximately $U/(2r)$, before other work is counted.

For broad source support, r can approach U and the factorized order need not be faster. Both orders evaluate the same orthogonal-CI action. An arithmetic planner may select between them without a memory-based admission test for the physical selected space. Exact applicability to arbitrary structures therefore does not mean that their costs are equal or that selecting one structure universally yields rank one. There is no claim of eliminating the exponential growth of the complete one-spin spaces.

8 Integral adjoints without a product carrier

Let the bra and ket amplitudes have exact factorizations

$$Y_\ell = L_\ell F_\ell^{\mathsf{T}}, \qquad Y_r = L_r F_r^{\mathsf{T}}, \qquad \mathscr{E} = \langle Y_\ell, \mathcal{H} Y_r \rangle_{\mathrm{F}}\,. \tag{40}$$

Their source counts r_ℓ and r_r need not be equal. At fixed factors, the derivative with respect to the Hamiltonian channel kernel is

$$\frac{\partial \mathscr{E}}{\partial K_{PQ}} = \left\langle \mathcal{E}_P Y_\ell, \mathcal{E}_Q Y_r \right\rangle_{\mathrm{F}} = G_{PQ}. \tag{41}$$

This uses the self-adjoint symmetric generators, but does not assume that the bra and ket, their source counts, or their spin spaces coincide.

First contract the unacted ket factors into the bra amplitude:

$$W_\alpha = Y_\ell F_r = L_\ell (F_\ell^{\mathsf{T}} F_r), \qquad W_\beta = Y_\ell^{\mathsf{T}} L_r = F_\ell (L_\ell^{\mathsf{T}} L_r). \tag{42}$$

For each channel P, form the two channel covectors

$$C_{\alpha P} = E_{\alpha P} W_\alpha + L_\ell \big(F_\ell^{\mathsf{T}} (E_{\beta P} F_r)\big), \tag{43}$$

$$C_{\beta P} = E_{\beta P} W_\beta + F_\ell \big(L_\ell^{\mathsf{T}} (E_{\alpha P} L_r)\big). \tag{44}$$

Their dimensions are $U_\alpha \times r_r$ and $U_\beta \times r_r$. Self-adjointness moves the crossed generator from the bra to the ket; for example, $(E_{\beta P} F_\ell)^{\mathsf{T}} F_r = F_\ell^{\mathsf{T}} (E_{\beta P} F_r)$. Thus no channel scatter on the bra source family is required, even when it contains many panels. The full kernel derivative is then

$$\boxed{G_{PQ} = \left\langle C_{\alpha P}, E_{\alpha Q} L_r \right\rangle_{\mathrm{F}} + \left\langle C_{\beta P}, E_{\beta Q} F_r \right\rangle_{\mathrm{F}}\,.} \tag{45}$$

To verify this identity, expand $\mathcal{E}_Q Y_r$ as $(E_{\alpha Q} L_r) F_r^{\mathsf{T}} + L_r (E_{\beta Q} F_r)^{\mathsf{T}}$ and move each unacted ket factor to the bra side of the Frobenius product. The resulting bra contractions are precisely Eq. (43). This is the reverse contraction of the same one-body channel graph used for sigma; no product-space one-body response or higher-order density tensor is formed.

Only $\operatorname{sym} G = (G + G^{\mathsf{T}})/2$ contributes to symmetric integral variations. Using the packed diagonal selector d_P and the effective one-electron matrix f defined above, Eq. (23) gives

$$\delta \mathscr{E} = \tfrac{1}{2} \langle \operatorname{sym} G, \delta V \rangle_{\mathrm{F}} + \sum_P \frac{[(\operatorname{sym} G) d]_P}{N}\, \delta f_P, \qquad \delta f_{ps} = \delta h_{ps} - \tfrac{1}{2} \sum_q \delta(pq|qs). \tag{46}$$

Unpacking the symmetric pair entries with half weight on each off-diagonal matrix element gives the full-matrix Frobenius gradients for h and V. The product and factor algorithms share this final kernel pullback in code; they differ only in the contraction order that generates G.

8.1 A bounded reverse sweep

For each ket source panel, stream the bra factors to accumulate W_α, W_β once. For each P-channel panel, scatter the ket and these weighted factors, then stream the bra base factors to accumulate the crossed contributions to $C_{\sigma P}$. Only after the complete bra has been consumed, stream the ket Q channels and add their Gram matrices to G. Putting the Q contraction inside the bra source loop would repeat the leading channel GEMM unnecessarily; that order is not used.

Let s_ℓ, s_r be the bra and ket panel widths and p the channel width. The required numerical workspace, measured in doubles, is bounded by

$$(U_\alpha + U_\beta)\big[s_\ell + 2 s_r + (2 + \mathbf{1}_{p<d}) s_r p\big] + s_\ell s_r p + \max(\omega_\ell s_\ell, \omega_r s_r). \tag{47}$$

It includes loaded bra/ket base factors, weighted bra factors, ket channel panels, channel covectors, and one reusable source moment matrix. When $p = d$, the ket P panel also supplies all Q channels; otherwise a separate Q panel is required. Here ω_ℓ, ω_r count temporary doubles per source column used by the respective factor loaders, excluding their output views. They vanish for the original exterior-factor provider; source generation is sequential, so the larger loading scratch, not their sum, enters the bound. The immutable metadata and the required returned $d \times d$ gradient are separate. Both source and channel widths can decrease to respect the budget, without changing the represented amplitudes or a derivative tolerance. Source and channel widths are chosen jointly by the repeated-work estimate in Section 8.4. A complete bra is retained when that schedule makes it worthwhile; otherwise the same equations stream its panels.

With complete channel panels, the leading dense arithmetic is

$$O\left((U_\alpha + U_\beta)[r_r d^2 + r_\ell r_r d]\right), \tag{48}$$

instead of product-carrier Gram work of order $U_\alpha U_\beta d^2$. The source-pair term is essential and is not assumed linear in rank. Self-adjointness permits exchanging the bra and ket before contraction, placing the smaller source family on the ket side. One-spin link traversals and panel regeneration also contribute; a smaller channel panel can repeat these traversals. No universal speedup is inferred from the leading dense terms alone.

For an exchange-parity reduction, a representative cannot replace both amplitudes without projection. A physical bra with parity η may, however, contract an unprojected ket representative Z when $Y_r = P_\eta Z$ and the integral derivative commutes with exchange: $\langle Y_\ell, \delta\mathcal{H} Z\rangle_{\mathrm{F}} = \langle Y_\ell, \delta\mathcal{H} Y_r\rangle_{\mathrm{F}}$. The caller must establish this identity; the common factor kernel does not infer a parity projection from equal spin dimensions.

8.2 Directional factors and the integral part of an HVP

Let R be the nonsingular upper-triangular orbital map in the exterior transforms and let δR be its direction. Define the spatial generator and its one-spin representation by

$$\kappa = \delta R R^{-1}, \qquad \mathcal{G}_\sigma(\kappa) = \sum_{pq} \kappa_{pq} a^\dagger_{p\sigma} a_{q\sigma}. \tag{49}$$

The derivative of an exterior representation obeys $\delta T_\sigma = \mathcal{G}_\sigma(\kappa) T_\sigma$. This generator differentiates the CI representation; it does not replace the physical nonorthogonal VB orbital coordinates or their finite update. For $Y = LF^{\mathsf{T}}$, with selected structure coefficients allowed to vary,

$$\boxed{\delta Y = (\mathcal{G}_\alpha(\kappa)L)F^{\mathsf{T}} + L(\mathcal{G}_\beta(\kappa)F)^{\mathsf{T}} + T_\alpha\, \delta X\, T_\beta^{\mathsf{T}}, \qquad \delta X = \mathcal{W}\, \delta\mathbf{C}.} \tag{50}$$

The structure map $\mathcal{W}$ and its exact support plan are fixed. The last term therefore uses the same support factorization with the structure-response coefficients; it is not a projection of either orbital derivative onto the original support. The two orbital terms generally leave that support and retain every contribution through their transformed spin factors.

Each term of Eq. (50) is another factor source. Concatenating them adds their outer products exactly, without SVD truncation or a closure assumption. A source panel loads its base factors and applies the oriented Slater–Condon generator to the required spin factor only. Unlike a Hamiltonian channel, κ need not be symmetric: replacing it by its symmetric part would give the wrong orbital derivative. The one-factor temporary uses at most $\max(U_\alpha, U_\beta)s$ doubles for panel width s; it is included in the loader workspace term of the preceding bound.

Write $\mathcal{J}(Y_\ell, Y_r)$ for the symmetric integral adjoint of $\langle Y_\ell, \mathcal{H} Y_r\rangle_{\mathrm{F}}$, including the shared K-kernel pullback. Since each physical integral variation is self-adjoint,

$$\delta\mathcal{J}(Y, Y) = 2\mathcal{J}(\delta Y, Y). \tag{51}$$

Thus the integral-gradient derivative needed in an orbital HVP can be contracted from the directional factor source directly, without constructing δY or any higher-order RDM for this contraction. If the accepted and directional factor counts are r and r_δ, the leading dense work is

$$O\left((U_\alpha + U_\beta)[\min(r, r_\delta)d^2 + r\, r_\delta d]\right), \tag{52}$$

plus the one-spin link traversals and source transforms. The count r_δ includes both orbital terms and any structure response, including the physical parity reconstruction when required. It must not be estimated from the accepted source count alone.

Equation (51) is the integral part of the HVP, not its complete orbital pullback. The derivative of the exterior-map adjoint, the full carrier residual, and the state-average gauge terms remain required. The present implementation keeps those terms while migrating the integral contraction to Eq. (51).

8.3 Oriented generator and residual contractions

Define the orbital matrix $\mathcal{D}(X, Y)$ by

$$[\mathcal{D}(X,Y)]_{pq} = \langle X, (a^\dagger_{p\alpha} a_{q\alpha} + a^\dagger_{p\beta} a_{q\beta}) Y\rangle_{\mathrm{F}}. \tag{53}$$

It is the coefficient of κ_{pq} in $\langle X, \mathcal{G}(\kappa)Y\rangle_{\mathrm{F}}$ and need not be symmetric. For $Y = L_r F_r^{\mathsf{T}}$, form only the weighted single-spin panels

$$W_\alpha = XF_r, \qquad W_\beta = X^{\mathsf{T}} L_r. \tag{54}$$

Then

$$[\mathcal{D}(X,Y)]_{pq} = \sum_j \langle (W_\alpha)_j, a^\dagger_{p\alpha} a_{q\alpha} (L_r)_j\rangle + \sum_j \langle (W_\beta)_j, a^\dagger_{p\beta} a_{q\beta} (F_r)_j\rangle. \tag{55}$$

If $X = L_\ell F_\ell^{\mathsf{T}}$, each weighted panel is obtained from source-pair moments, $XF_r = L_\ell (F_\ell^{\mathsf{T}} F_r)$ and $X^{\mathsf{T}} L_r = F_\ell (L_\ell^{\mathsf{T}} L_r)$. The final contractions traverse the oriented single-spin occupation/replacement links. They are not symmetric Hamiltonian-channel derivatives. For real coefficients, $\mathcal{D}(X,Y) = \mathcal{D}(Y,X)^{\mathsf{T}}$; this permits a transposed contraction order but not an unqualified interchange of sources.

For the residual $Z = (\mathcal{H} - E)X$, the required panels can also be obtained without Z or $\mathcal{H}X$. Using the same-spin actions D_σ and the ket channel mixtures

$$V_{\alpha P} = \sum_Q K_{PQ} E_{\alpha Q} L_r, \qquad V_{\beta P} = \sum_Q K_{PQ} E_{\beta Q} F_r, \tag{56}$$

the exact residual-weighted panels are

$$Z_\alpha = D_\alpha W_\alpha + X(D_\beta F_r) + 2\sum_P E_{\alpha P}(X V_{\beta P}) - E W_\alpha, \tag{57}$$

$$Z_\beta = D_\beta W_\beta + X^{\mathsf{T}}(D_\alpha L_r) + 2\sum_P E_{\beta P}(X^{\mathsf{T}} V_{\alpha P}) - E W_\beta. \tag{58}$$

Thus $Z_\alpha = ZF_r$ and $Z_\beta = Z^{\mathsf{T}} L_r$. Replacing W_α, W_β by Z_α, Z_β in Eq. (55) gives $\mathcal{D}((\mathcal{H} - E)X, Y)$. The spectator actions have been moved to the ket using self-adjointness of D_σ; the crossed term follows from Eq. (27). Self-adjointness of $\mathcal{H}$ does *not* justify exchanging the two sources here, because $\mathcal{H}$ generally does not commute with an oriented one-body generator.

The implementation reuses the factor/channel sweep twice: once for the weighted panels to produce $D_\sigma W_\sigma$, and once for the ket panels to produce the spectator and crossed terms. After same-spin gather, the ket's raw channel buffers are reused for $XV_{\beta P}$ and $X^{\mathsf{T}} V_{\alpha P}$, then consumed by the crossed gather. No separate channel scatter/mixing implementation or product-space residual is required.

With complete channel panels, the leading dense arithmetic is $O((U_\alpha + U_\beta)[r_r d^2 + r_\ell r_r d])$, plus the oriented single-spin link contractions. Without $\mathcal{H}$, it reduces to

$$O\left((U_\alpha + U_\beta) r_\ell r_r + r_r \sum_\sigma U_\sigma n_\sigma (n - n_\sigma + 1)\right). \tag{59}$$

Accepted-state transition densities use this estimate to choose between factor moments and product-space links. Their planner does not reuse the Hamiltonian-channel cost estimate.

A conservative numerical workspace bound for the residual contraction, in doubles, is

$$(U_\alpha + U_\beta)[s_\ell + (3p + 9)s_r] + s_\ell s_r \max(1, p) + \max(\omega_\ell s_\ell, \omega_r s_r) + (t + 1)n^2, \tag{60}$$

where t bounds the number of workers and the last term covers the small orbital-matrix reduction and its temporary result. The required returned $n \times n$ matrix and immutable metadata are separate. For a generator without $\mathcal{H}$, the channel sweep and residual outputs are absent, giving $(U_\alpha + U_\beta)(s_\ell + 2s_r) + s_\ell s_r$ plus the same loading and reduction terms. No per-worker spin/product array is allocated.

8.4 Joint source/channel scheduling

For source widths s_ℓ, s_r and channel width p, define

$$N_\ell = \lceil r_\ell / s_\ell \rceil, \qquad N_r = \lceil r_r / s_r \rceil, \qquad N_P = \lceil d / p \rceil, \qquad U = U_\alpha + U_\beta. \tag{61}$$

The leading dense channel and moment arithmetic does not depend on these widths, but source regeneration, link traversal and output traffic do. In particular, the number of loaded bra source columns is

$$C_\ell = \begin{cases} r_\ell, & s_\ell = r_\ell, \\ N_r(N_P + b) r_\ell, & s_\ell < r_\ell, \end{cases} \qquad b = \begin{cases} 1, & \text{integral adjoint}, \\ 2, & \text{residual generator}. \end{cases} \tag{62}$$

Here b counts the channel-independent moment sweeps. If generating a source column costs ℓ_ℓ nominal scalar operations and element touches, source work is $C_\ell \ell_\ell$. Each provider declares this work from its sparse writes, diagonal scales and exterior shear multiply–adds; directional providers add the oriented generator work. This metadata affects only the evaluation schedule, never the amplitudes, ranks or tolerances.

The single-replacement links are generated in ascending packed-channel order. For each larger orbital index, ascending lower indices with opposite occupation generate every link once. A channel tile then uses binary ranges in the existing link and occupation arrays. It does not scan all links and reject those outside the tile, nor allocate a string-by-channel offset table. Let $J = \sum_\sigma U_\sigma n_\sigma (n - n_\sigma + 1)$ count the single-spin occupation/replacement links. Across all source columns, matching-link visits scale as $r_r v(N_P) J$, where

$$v_I(N_P) = N_P + 1, \qquad v_R(N_P) = 2N_P + 3. \tag{63}$$

Without channel ranges these counts were respectively $N_P^2 + N_P$ and $2N_P^2 + 3N_P$. Binary lookup comparisons still repeat with those latter counts and are included through an upper comparison estimate B per complete single-spin row family. The complete-channel case needs no binary lookup penalty.

The schedule also counts repeated output reads and writes in the moment GEMMs, and repeated channel zero fills:

$$W_{\text{out}} = 2U r_r (d + b) N_\ell, \tag{64}$$

$$W_{0,I} = U r_r d (N_P + 1), \qquad W_{0,R} = 2U r_r d N_P. \tag{65}$$

Combining these terms gives the nominal variable-work estimate

$$\widehat{W} = C_\ell \ell_\ell + r_r \left[3v(N_P) J + s(N_P) B\right] + W_0 + W_{\text{out}}, \tag{66}$$

where $s_I = N_P^2 + N_P$, $s_R = 2N_P^2 + 3N_P$, and the factor three counts one matching-link visit plus its multiply and add. Terms invariant across the candidate schedules are omitted. The estimate combines scalar work and element traffic without fitted hardware weights; it is not a prediction of absolute wall time, which requires native measurement.

The planner searches feasible source and channel widths under the same memory bound. For each ket/channel choice it takes the largest feasible bra panel; ties use fewer moment panels. Partial-channel residual sweeps currently use one ket column because the shared forward workspace and moment buffers must use the same channel width. Full-channel sweeps allow wider ket panels. This is a workspace scheduling limit, not a restriction on the selected VB space.

9 Correspondence with the current implementation

The original column contraction corresponds to revision `a5004add`; Section 3.2 adds the minimum mixed-support decomposition. Paths below are relative to `src/`; this table records the implementation boundary rather than introducing physical assumptions.

Mathematical operation	Implementation
Original support and minimum source cover	`vbscf/structures/assembly/sources.cpp`
VB source generation, sampling and $\mathcal{W}^*$	`vbscf/structures/assembly/source_tiles.cpp` `apply_source_tiles()`
Common factor sweep and integral adjoint	`mcscf/ci/factors.cpp`
Generator and residual-weighted factor adjoints	`mcscf/ci/factor_generator.cpp`
Joint source/channel work plan	`mcscf/ci/factor_plan.cpp`
Hamiltonian channel kernel, Eq. (23)	`mcscf/ci/sigma.cpp` `build_hamiltonian_pair_kernel()`
Slater–Condon spin connections	`mcscf/ci/spin_channels.cpp`
Exterior transforms and causal adjoint rows	`vbscf/structures/orthogonal_ci/exterior_transform.cpp` `projection()` and `apply_adjoint_left_block()`
Independent complete-carrier comparisons	`../tests/vbscf/unit/test_structure_tiles.cpp`

Fixed-point structure Hamiltonian/overlap actions use the common factor sweep. Accepted-state integral gradients choose between the exact factor and product contractions by their arithmetic costs. Raw state coefficients are retained in structure space. The orthogonal state, its full carrier residual, and generator-response consumers still require migration. Directional integral adjoints use Eq. (51) when the factor order has less estimated arithmetic. A selected-space eigenvector need not be stationary on the full carrier: its residual contribution to the exterior-transform gradient is retained. Accepted-state transition densities use Eq. (55) when advantageous. The residual-panel kernel of Eq. (57) is available and independently tested; accepted-state residual storage is not yet removed from the remaining consumers.

Main result. Equations (27)–(33) give the exact selected-space restriction through spin factors and Slater–Condon actions, without a closure assumption or a complete transformed product array.